



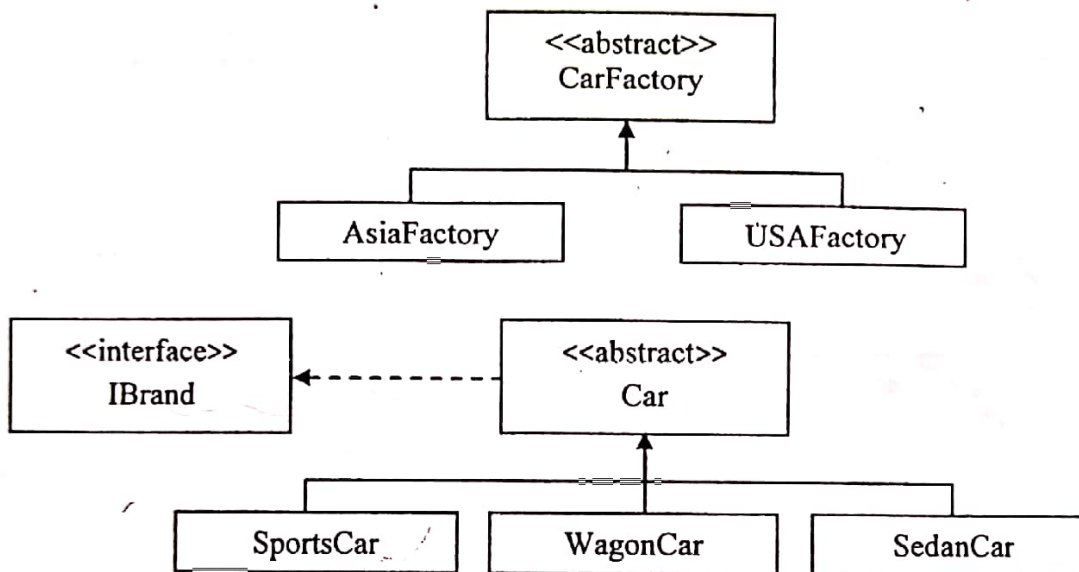
UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER II) - 2013/2014
CS105-OBJECT ORIENTED PROGRAMMING PRACTICALS (1 Credit)

Answer ALL Questions

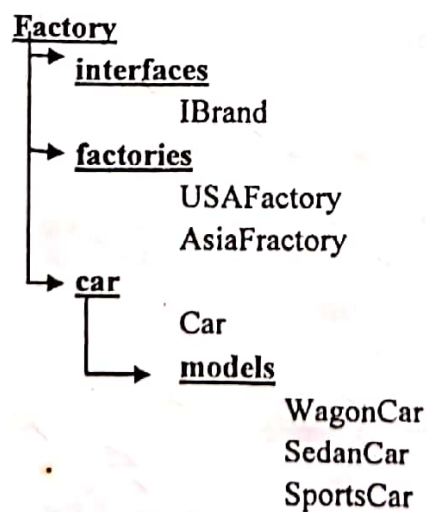
Time Allowed: Three hours

- Driver class files should be saved as xxxxxxQ1.java and xxxxxxQ2.java in your home directory (i.e. in H:\), where xxxxxx denotes your registration number.
- Add comments to the code where necessary.

Q1) Implement the classes shown in the following diagram for a car manufacturing system.



Include the above classes in the following package structure.



The attributes and methods of the classes are given below.

<<abstract>> Car
public String model public String location public Date manufacturedDate public double price; public String color
public Car(String m, String l) public void modify(String c) public abstract void assignCarPrice()

<<interface>> IBrand
public void displayBrand()
WagonCar
Public WagonCar (String m,String l)

SportsCar
Public SportsCar (String m,String l)

SedanCar
Public SedanCar (String m,String l)

<<abstract>> CarFactory
public String city
Public CarFactory(String c) public abstract Car buildCar(String m)

USAFactory
Public USAFactory()

AsiaFactory
Public AsiaFactory ()

- Implement the Car class *constructor* to initialize model, location and manufactured date. Manufactured date should be the current date. The *modify* method must be used to set a color to the car.
- The *assignCarPrice* should set the price as follows.

Model	Price (\$)
Wagon	15,000
Sedan	290,000
Sports	550,000

- ii) The *displayBrand* method should be implemented to display car model, location and its brand. The brand of each car is given below.

Model	Brand
Wagon	Ford Fiesta
Sedan	Rolls-Royce Ghost
Sports	Lamborghini Aventador

- iii) Implement both "AsiaFactory" and "USAFactory" constructors to assign appropriate cities as follows for each factory.

USAFactory- New York
AsiaFactory - Tokyo

- iv) Implement the *buildCar* method in both "AsiaFactory" and "USAFactory" to create car objects.

AsiaFactory can only build Wagon and Sedancars
USAFactory can only build Sedan and Sports cars.

- v) Create an array in the driver class to store following car objects

A Sedan car made in AsiaFactory
A Sports car made in USAFactory
A Sedan car made in USAFactory
A Wagon car made in AsiaFactory

- vi) Perform *displayBrand* and *assignCarPrice* methods on all the above created car objects.

(50 marks)

Q2) Inventory details of a store are given in the 'itemData.txt' file. Each entry contains details of a single item. The data is given in the following format.

ItemCode, Description, Quantity, Retail Price, Sales Price

Note: Price values are given for a single item.

- Create a class 'Item' to store the details for a particular item and the profit of the store.
- Create a driver class to perform the following tasks.
 - Check whether the file exists in 'H:\files\ItemData.txt'. If the file does not exist or/and file is empty, prompt a proper error message.
 - If the file exists and not empty, read the data and store them in a vector of 'Item' objects.

- c) Create a method to calculate the **expected profit** of the store. The profit which is calculated as follows needs to be Stored.

$$\sum_{\text{forall the items}} (\text{Salesprice} - \text{Retailprice}) * \text{Quantity}$$

- d) Create a method to generate the bill considering the following scenario. You should perform the following,

- i. The **item code** should be passed to the method. If the item exists allow the user to enter the purchased quantity of the item.
- ii. If the purchased quantity is **greater** than the existing quantity in the store, then calculate and display the maximum possible quantity of items that can be purchased. Proceed ahead with that quantity and calculate the **amount**.

- iii. Discounts can be given as follows,

If the amount is,

Over 1000 - Discount is Rs: 100/=

Over 3000 - Discount is 5% of the amount

- iv. Display the **total amount** to be paid and then allow the user to enter the amount that the customer pays and calculate the balance.

2

Total amount = amount - discount

- v. Write the bill as follows to a separate file called 'myBill.txt'.

Note: You should display the current date as Bill Date.

***** Chamara Stores *****

Bill No:001

Bill Date : Fri Nov 28 00:14:32 IST 2014

Item Code	Desc	Price	Qty	Amount
R001	Shampoo Bottle	530.00	2	1060.00

Discount :				100.00
TOTAL :				960.00
Cash :				1000.00
Balance :				40.00

3.

***** Thank You, Come Again *****

(50 marks)